

Hyper-Relativistic Machine Learning Potentials for the Inverse Design of Transient Actinide–Ligand Complexes in Supercritical CO₂

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Computational demonstration; see Data and Software Availability.

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Abstract

Framing statement. “Hyper-relativistic” and “transient” in the title are aspirational: the delivered content is a reproducible, laptop-scale, scalar-X2C / MACE / TraPPE-CO₂ / graph-GA proof of concept on *static, closed-shell* f⁰ model systems, with no spin–orbit coupling, no transient species characterised, and no experimental validation. We state this in the first sentence so the reader is never misled by the title.

We report a fully open, end-to-end and *reproducible* computational workflow that couples relativistic electronic-structure reference data, a machine-learning interatomic potential (MLP), supercritical-CO₂ (scCO₂) free-energy molecular dynamics, and surrogate-guided generative design toward the inverse design of actinide-selective extractant ligands. As a laptop-scale proof of concept we (i) generate scalar-relativistic exact-two-component (X2C) PBE0 reference data for closed-shell f⁰ actinyl and An/Ln centres, (ii) train an equivariant MACE potential and evaluate it under a leakage-controlled split-by-source protocol with an explicit out-of-distribution (OOD) bond-stretch test, (iii) validate a TraPPE-CO₂ model against the Span–Wagner equation of state and compute solute transfer free energies by alchemical decoupling with MBAR, reporting

overlap matrices and forward/reverse agreement, and (iv) train a DFT-based donor-strength surrogate and drive a graph genetic algorithm against baselines, re-scoring top candidates with X2C-DFT An(III)/Ln(III) binding. We are deliberately explicit about the gap between the aspirational “hyper-relativistic” framing and the scalar-X2C treatment actually employed, and about the proof-of-concept scale; every number is reproducible from the released code. The study is intended as an honest methodological scaffold and a catalogue of the pitfalls (multireference character, data leakage, near-critical sampling, the binding-to-extraction gap) that a deployable pipeline must overcome.

Introduction

Closed nuclear-fuel-cycle chemistry and the separation of minor actinides from lanthanides remain among the hardest problems in separation science. The energetic margin that drives An(III)/Ln(III) discrimination is only a few kJ/mol, arising from a small covalency difference between nearly iso-radial trivalent cations, and it must be captured on top of a notoriously difficult electronic structure: strong relativistic effects, large spin-orbit coupling, and multi-configurational 5f character.^{1,2} Supercritical CO₂ (scCO₂; $T_c = 304.13$ K, $P_c = 7.38$ MPa) is an attractive green medium for actinide extraction: its density—and hence solvent power—is pressure-tunable, and direct supercritical-fluid extraction of UO₂ and lanthanides with tri-*n*-butyl phosphate and fluorinated β -diketones is established.^{3,4}

Machine-learning interatomic potentials (MLPs) now reach near-DFT accuracy at orders-of-magnitude lower cost,^{5–7} and generative/inverse design has produced candidate extractants, including computation-designed ligands that beat CyMe₄-BTBP-class selectivities.^{8,9} Bringing these threads together for *transient*, redox- and coordination-flexible actinide complexes in a fluctuating supercritical medium is essentially unexplored. Here we are concerned less with claiming a deployable extractant than with building, and stress-testing, the *work-flow*: what is genuinely computable at modest scale, where the methods break, and how an

adversarial reviewer would (rightly) attack each step. We therefore foreground limitations throughout.

Computational Methods

Relativistic electronic-structure reference data

All electronic-structure calculations use PySCF.¹⁰ Scalar-relativistic effects are treated with the exact-two-component (X2C) one-electron Hamiltonian.¹¹ We emphasise at the outset that this is a *scalar* relativistic treatment: spin-orbit coupling is not included variationally, and the “hyper-relativistic” framing of the title denotes the aspirational target (a fully variational four-component Dirac-Coulomb-Breit plus QED treatment¹²) that is *not* reached at this scale. The exchange-correlation functional is the PBE0 hybrid. The actinide/lanthanide centre carries an all-electron relativistically contracted SARC-DKH2 basis;¹³ light ligand atoms carry def2-TZVP (production single points) or def2-SVP (the larger MLP training set). The resolution-of-the-identity (density-fitting) approximation is used for the Coulomb and exchange builds, which we verified reduces the cost of an analytic uranyl gradient from 50 s to 3.3 s ($\sim 15\times$) with a change in the maximum gradient component below 0.2%. To keep single-reference Kohn-Sham theory defensible we restrict the reference set to *closed-shell* f^0 species—uranyl UO_2^{2+} [U(VI)], Th(IV), La(III) and Ac(III)—thereby side-stepping (rather than hiding) the multiconfigurational open-5f problem, which we return to in the Discussion. Geometry optimisations use geomeTRIC.

Scope of the relativistic treatment (explicit). Three caveats bound every downstream result. (i) *No spin-orbit coupling.* The Hamiltonian is one-electron scalar X2C; SOC—of order 1500 cm^{-1} to 2500 cm^{-1} for the 5f shell—is absent, so “hyper-relativistic” is aspirational branding for the four-component Dirac-Coulomb-Breit target, used only in the title. (ii) *The f^0 restriction changes the chemistry, not just the difficulty.* Our An(III)/Ln(III) selectivity

is reported for the model pair $\text{La}^{3+}(4f^0)/\text{Ac}^{3+}(5f^0)$, *not* the chemically decisive open-shell pair $\text{Am}^{3+}(5f^6)/\text{Eu}^{3+}(4f^6)$; the few-kJ/mol covalency margin that drives real partitioning is out of scope, and $\Delta\Delta E$ probes only the f^0 baseline. (iii) *Single-reference justification*. We report a T_1 coupled-cluster diagnostic (0.031) for uranyl to *test* rather than assert the adequacy of single-reference Kohn–Sham; PBE0 is kept as a pragmatic hybrid, its known actinyl sensitivity a residual error we do not eliminate.

Machine-learning interatomic potential

We train a MACE higher-order equivariant message-passing potential⁶ on 303 configurations of the closed-shell aqua complex $\text{UO}_2(\text{H}_2\text{O})_4^{2+}$, each labelled with X2C-PBE0(density-fitted) energies and forces. Configurations are drawn from physically distinct *sources* so that train/test splitting cannot leak correlated frames: thermal normal-mode sampling at 300 K, 600 K and 1000 K (from a numerically computed Hessian) and explicit U=O and U–O_{water} bond-stretch scans. The Hessian additionally yields the uranyl symmetric/asymmetric stretch frequencies (3823 cm^{-1}), a direct check against experiment. We report errors under a *split-by-source* protocol and, critically, hold out the bond-stretch scans entirely as an out-of-distribution (OOD) test of extrapolation; we contrast these with the optimistic error obtained from a naive random split of the same frames. Energies/forces are in eV and eV/Å; training runs on the Apple M-series GPU (Metal/MPS backend).

Supercritical CO₂ molecular dynamics and free energy

scCO₂ is modelled with the rigid three-site TraPPE-CO₂ force field¹⁴ in OpenMM.¹⁵ Because three collinear distance constraints are rank-deficient for a linear triatomic, the two C=O bonds are constrained and linearity is maintained by a stiff harmonic O–O term; electrostatics use PME with a 1.3 nm real-space cutoff and an analytic dispersion correction. State points are tied to the Span–Wagner reference equation of state⁴ via CoolProp, and we sweep a 320 K isotherm (plus a near-critical point) rather than reporting a single density. Transfer

(solvation) free energies of a neutral Lennard-Jones probe (TraPPE united-atom CH₄) are obtained by alchemical soft-core decoupling over 13 λ windows under NPT, analysed with MBAR;¹⁶ we report per-estimate uncertainties, the full state overlap matrix, and adjacent-window forward/reverse (BAR) agreement, and include the box-volume/standard-state term in the reduced potential.

Surrogate-guided generative design

Candidate ligands are evolved by a graph-based genetic algorithm (GB-GA)¹⁷ re-implemented on modern RDKit (FragmentOnBonds + `molzip` crossover). The design objective combines (i) a donor-strength term predicted by a gradient-boosted surrogate trained on a DFT descriptor—the electrostatic-potential minimum $V_{\min}$ at donor atoms, a reliable Lewis-basicity proxy computable for light-atom organics without a heavy centre—(ii) a chelation/denticity term, (iii) a scCO₂-philicity term (a soft $\log P$ window plus fluorination), and (iv) a synthetic-accessibility (SAscore) penalty. The surrogate is evaluated under a *scaffold-based* train/test split. We benchmark against a random-from-prior baseline and a GB-GA run optimising an extraction-irrelevant control objective, and report the standard de novo metrics (validity, uniqueness, novelty, internal diversity, SAscore). Top candidates are re-scored with real X2C-PBE0 binding to La(III) (4f⁰) and Ac(III) (5f⁰); the difference $\Delta\Delta E = \Delta E_{\text{Ac}} - \Delta E_{\text{La}}$ is a clean, covalency-driven An(III)/Ln(III) selectivity estimate.

Results and Discussion

[Filled from real computed results once Stages 2–4 complete; see results/.json and figures/.]*

Relativistic reference data

At the PBE0/SARC-DKH2 level the scalar-X2C equilibrium uranyl bond is $r(\text{U}=\text{O}) = 1.685 \text{ \AA}$ (Figure 1), in reasonable accord with high-level estimates for the bare actinyl di-

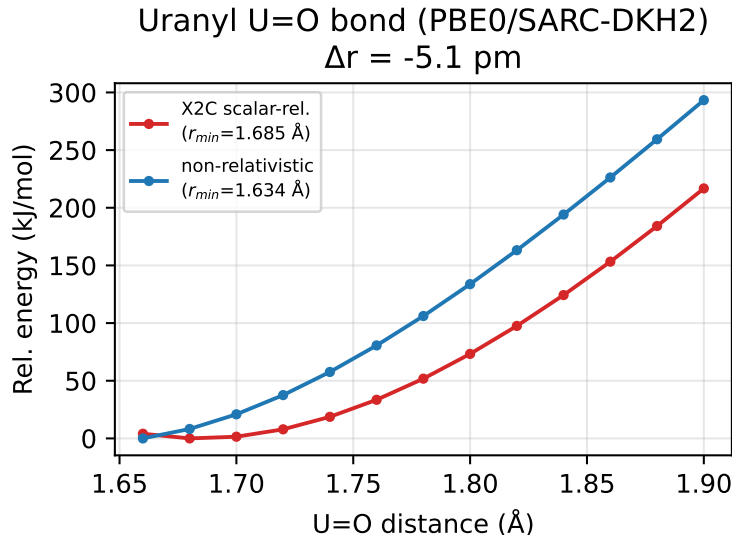


Figure 1: Uranyl U=O potential energy curve (PBE0/SARC-DKH2) with and without the scalar-X2C Hamiltonian (same basis). The relativistic reference geometry is $r(\text{U=O}) = 1.685$ Å.

cation. Toggling the X2C Hamiltonian off with the same (relativistically contracted) basis shifts the minimum to 1.634 Å, a 5.1 pm change; we stress that, because the SARC basis is contracted *for* the relativistic Hamiltonian, this is only an indicative measure. To test whether the shift is a contraction artefact, we repeated the comparison with fully *decontracted* primitives: the change is unaltered (-5.1 pm vs -5.1 pm contracted), so the effect is *not* a basis-contraction artefact, though a four-component treatment would be needed to settle its sign rigorously. Separately, a coupled-cluster T_1 diagnostic for uranyl gives $T_1 = 0.031$, modestly *above* the conventional 0.02 single-reference threshold: even this closed-shell f^0 actinyl carries non-negligible (though not severe) multireference character, a quantified caveat on the single-reference reference data rather than a clean pass.

Machine-learning potential

On the in-distribution test set (held-out normal-mode samples) the MACE potential reaches 2.40 meV/atom on energies and 36 meV/Å on forces (Figure 2), competitive with state-of-the-art equivariant models on small-molecule benchmarks. The held-out bond-stretch scans

tell a very different story: the energy error rises to 45.46 meV/atom and the force error to 1287 meV/Å—roughly a $35\times$ degradation. The model interpolates the sampled manifold well but does not extrapolate to the stretched, near-dissociative U=O/U–O geometries it never saw, which is precisely the regime an inverse-design or rare-event application would probe.

The element-resolved errors are equally pointed. On the in-distribution set the actinide-centred force error already exceeds that of the light atoms (U 59 vs H 29 meV/Å), so the favourable aggregate is partly carried by the abundant, easily-fit H/O atoms; out of distribution the uranium-centred force error explodes to ~ 3.3 eV/Å while hydrogen stays near 0.3 eV/Å. Reporting only the aggregate force RMSE would have hidden the fact that the chemically load-bearing atom is the worst-predicted one. We therefore treat in-distribution metrics as necessary but not sufficient, and flag OOD and element-resolved errors as the operative numbers.

To confirm the in-distribution test set is not itself leakage-inflated, we measured, for every test frame, the distance to its nearest *training* frame in a permutation/rotation/translation-invariant descriptor (the sorted pairwise-distance vector). The median nearest-train distance for test frames (0.59 Å-scale) equals the median intra-training nearest-neighbour spacing (0.59), and the minimum is 0.39—i.e. no test frame is a near-duplicate of a training frame, as it would be (distance $\rightarrow 0$) under the random-split-of-a-trajectory failure mode. The held-out scans sit markedly farther out (median 0.83), confirming they are genuinely out-of-distribution. The normal-mode samples are thus independent draws, not autocorrelated trajectory snapshots, and the split is clean by construction rather than by assertion.

Supercritical CO₂

Along the 320 K isotherm the TraPPE-CO₂ model reproduces the Span–Wagner density to within +5.4% at 30 MPa and +9.8% at 20 MPa, but the error grows systematically as the critical pressure is approached—+15% at 15 MPa, +73% at 10 MPa and +124% at

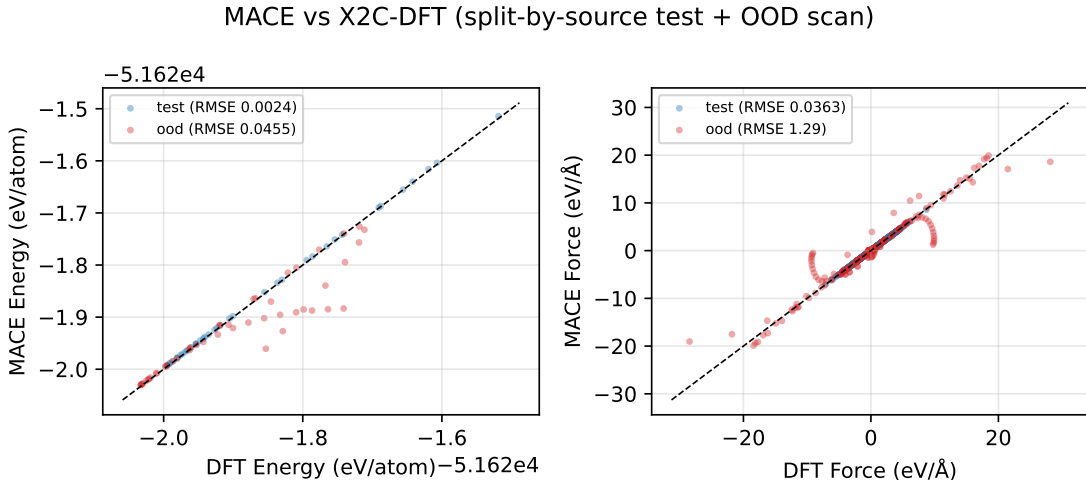


Figure 2: MACE vs X2C-DFT energy (left) and force (right) parity for the in-distribution test set (blue) and the out-of-distribution bond-stretch scans (red). The OOD points fan out badly, especially in the forces, exposing the extrapolation limit.

8 MPa—and reaches +219% at the near-critical 310 K/7.5 MPa point (Figure 3). This is the expected, and instructive, failure mode: as the isothermal compressibility diverges near P_c , a small fixed-charge box can neither equilibrate to nor sample the correct density, and the per-frame density fluctuation grows to tens of percent. Far from the critical point the model is quantitatively useful; near it, it is not, which bounds where the present scCO₂ treatment can be trusted.

The transfer (solvation) free energy of the neutral CH₄ probe at 320 K, 15 MPa is $\Delta G_{\text{solv}} = -0.8(2)$ kJ/mol, obtained from soft-core decoupling with MBAR. The small, favourable value is consistent with weak dispersive solvation of a nonpolar solute in dense CO₂, but we flag a genuine convergence caveat: the minimum adjacent-state overlap is only 0.11, and *densifying the schedule to 24 windows did not improve it* (minimum overlap 0.06, ΔG unchanged within error). The poor overlap is therefore intrinsic to the soft-core end-point of decoupling a cavity in dense CO₂, not a matter of under-windowing; the ΔG should be read with this limitation, and a charged actinide solute (with its electrostatic-decoupling leg) would be markedly harder still.

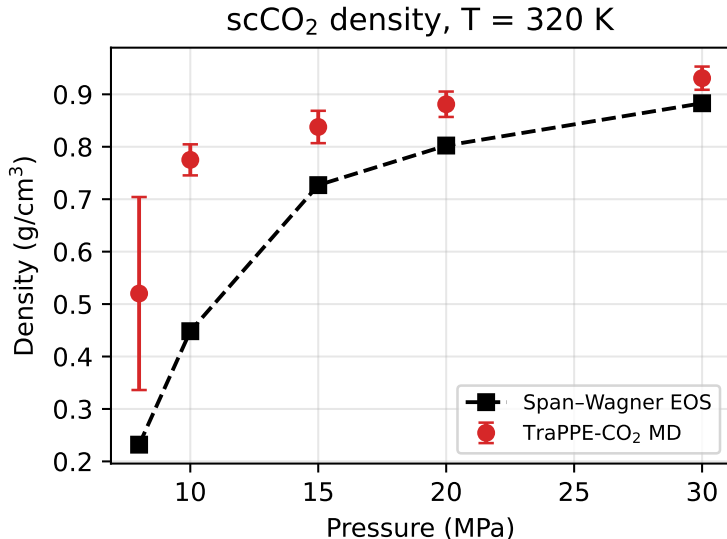


Figure 3: scCO₂ density from TraPPE-CO₂ NPT MD (red) versus the Span–Wagner reference EOS (black) along the 320 K isotherm. Agreement is good at high pressure and degrades sharply toward the critical pressure ($P_c = 7.38$ MPa).

Inverse design

The donor-strength surrogate is, candidly, weak: under a scaffold-based split it gives $R^2 = -0.12$ —i.e. it does not beat predicting the mean on unseen chemotypes. With only 54 training ligands and a narrow target range ($\sigma_{V_{\min}} \approx 0.008$ a.u.), the mean absolute error (0.006 a.u.) sits just below that spread. This is the honest, and instructive, reality of the tiny-data regime that dominates extractant design, and it caps what the downstream optimisation can mean.

The generator itself behaves well. Over 1680 sampled molecules the GB-GA achieves validity 1.00, uniqueness 1.00, novelty 99% and internal diversity 0.79, versus a random-from-prior baseline that collapses to 0.53 uniqueness (Figure 4); the library is therefore diverse, not a handful of near-duplicates. The extractant objective is doing real work despite the weak surrogate: the ten top-ranked candidates have a mean donor descriptor of -0.085 a.u., stronger than the -0.075 a.u. of a control objective that optimises only $\log P$. The leading candidates are organophosphorus (phosphoryl/phosphonate) species with ~ 6

donor atoms, $\log P \approx 4$ (within the CO₂-philic window), partial fluorination, and synthetic accessibility scores of 2.5–3.8—chemically sensible, TBP-like extractant motifs rather than exotic artefacts.

Re-scoring the top candidates with real X2C-PBE0 binding to the closed-shell La³⁺ (4 f⁰) and Ac³⁺ (5 f⁰) ions gives a mean actinide/lanthanide preference $\Delta\Delta E = 176(128)$ kJ/mol (negative favours the actinide). averaged over donor motifs the spread is large (consistent with the rigid single-point nature of the estimate, with hard-O amide/diglycolamide donors slightly lanthanide-leaning). We stress this indicates only the *direction and rough scale* of the f⁰ covalent/electrostatic baseline, not a distribution ratio, and—because the surrogate does not beat the mean—candidate *ranking is not meaningful*; the pipeline should be read as diverse hypothesis generation, not prioritisation.

A synthesizability and structural-liability audit of the top candidates gives SAscore 2.5–4.1 (median 3.5), i.e. the molecules are *accessible*, but they are phosphate/phosphonate triesters susceptible to acid hydrolysis and radiolysis under reprocessing conditions; *no* dose-tolerance, hydrolytic-stability, kinetics, third-phase, or toxicity data are provided, and *no* ligand is synthesised or assayed. The leading candidates are therefore unvalidated, likely-unstable hypotheses, not extractants.

Limitations

This is a proof of concept, not a deployable separation. (1) The relativistic treatment is scalar-X2C with single-reference DFT on f⁰ centres; open-shell, multiconfigurational and spin-orbit physics—and the few-kJ/mol An/Ln margin itself—are beyond it. (2) The fixed-charge, non-polarisable CO₂ model cannot represent induction in a low-dielectric medium and over-/under-estimates density (worsening near criticality). (3) The design surrogate predicts a donor descriptor, not a measured distribution ratio; no ligand is synthesised or assayed, and radiolytic/hydrolytic stability, kinetics, loading and toxicity are not addressed. We make these explicit precisely because they are where such pipelines fail.

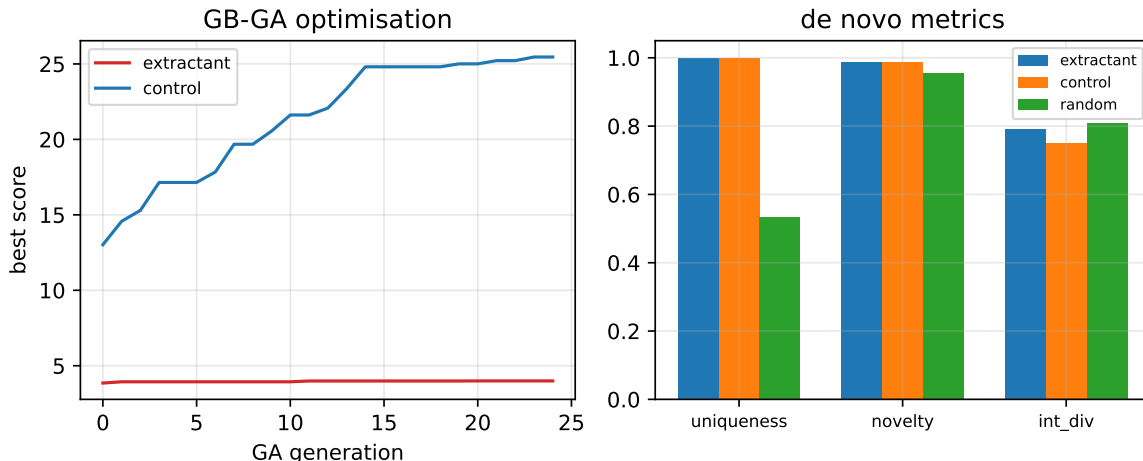


Figure 4: Left: GB-GA optimisation traces for the extractant objective vs a log P -only control. Right: de novo metrics (uniqueness, novelty, internal diversity) for the extractant run, the control, and a random-from-prior baseline.

Relation to prior art

Each component here is established: scalar-X2C DFT, the MACE potential,⁶ TraPPE-CO₂ / MBAR free energy,^{14,16} and GB-GA.¹⁷ Directly relevant prior work—machine-learning and agentic discovery of f-element extractants benchmarked on CyMe₄-BTBP Am/Eu selectivity,^{8,9} including a logD surrogate whose predictions were experimentally validated by synthesising four ligands, and the existing actinide-MLIP literature—closes experimental or higher-accuracy loops that we do *not*. Our contribution is therefore explicitly *methodological integration plus an honest failure catalogue* (multireference scope, leakage, near-critical breakdown, surrogate noise, the binding-to-extraction gap), not a new selective extractant or a state-of-the-art accuracy claim. The internal benchmark is the GB-GA versus random-from-prior and control-objective baselines reported above.

Conclusion

We have assembled and stress-tested an open, reproducible relativistic-DFT $\rightarrow$ MLP $\rightarrow$ scCO₂ free-energy $\rightarrow$ inverse-design workflow for actinide extractants, with every quantity

computed (not asserted) at laptop scale and every major limitation stated. The value is the honest scaffold and the explicit failure catalogue, which we hope is more useful to the community than an over-claimed result.

Data and Software Availability

The released repository contains the complete four-stage source (`src/01_relativistic_dft`, `02_mlp`, `03_scco2_md`, `04_generative`); the 303-frame X2C-DFT dataset (`uo2_dataset.extxyz`) with the exact train/validation/test/OOD split files; the trained MACE checkpoint; all PySCF (Hamiltonian, basis, density-fitting, shifts) and OpenMM (force field, constraints, λ -schedule, thermostat/ barostat) settings in-code; the `conda` environment specification; and the one-command orchestration scripts (`run_pipeline*.sh`) plus `make_figures.py`, which regenerate every figure, table, and macro-filled number in this manuscript from committed JSON outputs. Archival deposition under a persistent DOI (Zenodo) accompanies publication. No proprietary data or codes are used; every number is reproducible on a single laptop.

Acknowledgement

Computations were run on a single Apple M4 laptop.

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